

# Asmita Pal

Address: 1415 Engineering Drive, Madison, WI 53706

Website: [asmita-pal.github.io](https://asmita-pal.github.io) | LinkedIn : [linkedin.com/in/asmita-pal/](https://linkedin.com/in/asmita-pal/)

Email: [asmita.pal@wisc.edu](mailto:asmita.pal@wisc.edu) | Phone: 480-516-3041

---

## Summary

PhD candidate in Computer Architecture specializing in performance modeling, memory subsystems and microarchitectural security. Currently seeking research or engineering roles at the intersection of architecture and security.

---

## Education

### PhD in Computer Engineering

University of Wisconsin - Madison

August 2018 - August 2025

### MS in Computer Engineering

Utah State University

August 2016 - August 2018

### B.Tech. in Electronics and Instrumentation Engineering

West Bengal University of Technology, India

July 2010 - June 2014

---

## Research and Work Experience

STACS Lab, University of Wisconsin - Madison

MADISON, WI

### Graduate Research Assistant

August 2018 – Present

*Advisor* : Joshua San Miguel

### Side-channel attacks for mesh-based Network-on-Chips(NOC)

- Identified contention for Network-on-Chips for mesh interconnects using traffic patterns on Garnet in gem5.
- Developed temporal partitioning techniques inspired by SurfNoC to mitigate interconnect-level information leakage in SoCs.

### Fully Homomorphic Encryption

- Developing analytical models for performance prediction of FHE-based applications. Followed by guided parameterization for ML-based predictors.
- Analyzed side-channels in applications encrypted using FHE (Fully Homomorphic Encryption), on the IBM HELayers framework.
- Identified spatial and temporal memory access patterns in encrypted queries that leak sensitive information.
- Demonstrated 90% accuracy in distinguishing database entries using ML classifiers, revealing a critical privacy vulnerability in secure computation.

### Camouflage : Privacy-preserving Traces

- Addressed challenges in sharing confidential workloads for architectural research by balancing trace utility and privacy.
- Characterized utility of a trace as a function of program semantics that influence program behavior, as captured in memory traces.
- To ensure input privacy, developed a threat model based on input indistinguishability and show that the average security loss is 7.8% for obfuscated traces.
- Tested camouflaged encryption traces against secret key recovery.
- Proposed a trace-based obfuscation mechanism for workloads from SPEC2017, GAPBS and PERFECT Benchmark suites. Validated these traces against prefetchers and replacement policies on ChampSim.

### As-Is Approximate Computing

- Applied reduced precision, random input sampling and tree-based sampling techniques for an approximate compute model, which is interruptive.
- Simulated such approximate applications (PERFECT and GAPBS Benchmarks) on a speculation-based architectural simulator for studying accuracy-runtime trade-offs.

### Prefetching using Large Language Models

- Applied NLP-inspired techniques to memory access traces for learning temporal and spatial locality and predict addresses to prefetch.
- Compared hardware prefetcher vs LLM-predicted addresses to understand prefetcher accuracy and coverage.

## Course Projects

### Digit Recognition on FPGA using Multilayer Perceptron

- Designed processor, memory and accelerator modules to enable storage of a pre-trained model and offload computations to accelerator.
- Captured image and predicted image digit on the 7 segment display.

### Precise encoding in Stochastic Computing for Neural Networks

- Proposed a new encoding scheme for Stochastic Numbers and implemented this new encoding format in MAC module of a small-scale Neural Network.
- Analyzed the energy-accuracy trade-off for LeNet300-100 and observed up to 5X energy improvement.

## AMD Research

BOSTON, UNITED STATES

### Research Intern

June 2022 – Sept 2022

- Characterized last-level cache (LLC) traffic patterns for AI and HPC workloads on multicore CPU SoCs.
- Investigated heuristics and machine learning-based methods for improved cache management.
- Explored reuse-distance metrics to guide LLC replacement policies.
- Pending US Patent for reuse distance-based estimations for cache management.

## AMD Research

BOSTON, UNITED STATES

### Research Intern

September 2020 – December 2020

- Collected memory and last level cache statistics using hardware performance counters.
- Used metrics based on software-hardware interaction and analyzed different kernels present in BERT to determine potential areas of acceleration.
- Generated kernel-specific traces to study performance and sensitivity to memory latency/bandwidth of this application on simulator.

## Bridge Lab, Utah State University

UTAH, UNITED STATES

### Graduate Research Assistant

August 2016 – July 2018

### Design of a Process-Variation Aware Low Power GPU Supervisor : Koushik Chakraborty, Sanghamitra Roy

- Explored the role of process variation on register files for a GPU operating at Near Threshold Voltages.
- Analyzed the performance variation in GPGPU applications from AMD APP SDK suite, with varying register latency.
- Proposed a low overhead technique to overcome the effects of process variation on register latency. Exhibited ~36% reduction in GPU energy consumption with marginal area and power overheads.

## Course Projects

### Speaker Diarization using Deep Neural Networks

- Extracted features from a conversation and used Deep Neural Networks to separate an audio signal according to speaker identity.
- Compared accuracy of the speaker diarization framework, using different deep neural networks (DNN) and convolutional neural networks (CNN).

### Critical Thread Predictor

- Studied the statistical correlation of thread criticality metrics based on cache-misses and synchronization bottlenecks.
- Analyzed performance numbers for Splash2 Benchmarks on multi-core Sniper simulator.

## Infosys Technologies Limited

PUNE, INDIA

### System Engineer

July 2014 – August 2016

- Developed an automation framework for hardware verification, streamlining debugging processes and reducing manual verification time, to enhance team productivity.
- Managed client deadlines and service delivery, coordinating with cross-functional teams.

---

## Publications

- **Asmita Pal**, Keerthana Desai, Rahul Chatterjee, Joshua San Miguel: **Camouflage: Utility-Aware Obfuscation for Accurate Simulation of Sensitive Program Traces**, published in ACM Trans. Archit. Code Optim. 21, 2, Article 36 (June 2024), 23 pages.
- Mitali Soni, **Asmita Pal**, Joshua San Miguel: **As-Is Approximate Computing**, published in ACM Trans. Archit. Code Optim. 20, 1, Article 3 (March 2023), 26 pages.
- **Asmita Pal**, Karthik Swaminathan, Subhankar Pal, Joshua San Miguel: **Characterizing Memory side channels in FHE applications** presented in DISCC Workshop (held in conjunction with MICRO'22)

- Pramesh Pandey, **Asmita Pal**, Koushik Chakraborty, Sanghamitra Roy: **Reliability and Uniformity Enhancement in 8T-SRAM based PUFs operating at NTC**, published in IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED) 2018.
- **Asmita Pal**, Aatreyi Bal, Koushik Chakraborty, Sanghamitra Roy: **Split Latency Allocator: Process Variation-Aware Register Access Latency Boost in a Near-Threshold Graphics Processing Unit**, published in J. Low Power Electronics 13(3): 419-427 (2017).

---

## Awards

- Winner of N+1 Institute Google Reverse Pitch Competition - (AI Sustainability) at UW-Madison.
- Sarah and David Epstein Teaching Fellowship
- Student Research Grant [Awarded by Office of Diversity, Inclusion and Funding at **UW-Madison**]
- ISCA 2022 Student Travel Award
- Anita Borg Grace Hopper Celebration 2019 Scholarship
- Wisconsin Distinguished Fellow - Schneider
- Rajendra Prasad and Yasoda Mani Grandhi Fellow

---

## Research Interests & Technical Skills

**Research Interests :** Computer Architecture, Microarchitecture, Memory Subsystem, Performance Modelling, Approximate Computing, Privacy-preserving Traces, Fully Homomorphic Encryption, Interconnects, Side-channel Attacks, AI Sustainability

**Programming Languages :** C++, Python, Bash, Verilog

**Architectural Tools :** ChampSim, PinTool, Perf, likwid, Gem5, Garnet, Sniper

**Security Libraries :** HELayers, OpenFHE

---

## Professional Service

### Invited Talks

- [Town Hall](#) - University of Wisconsin - Madison, Dept. of Electrical and Computer Engineering, February 2025
- [Computer Architecture Coordination Seminar Series](#) - University of Central Florida, November 2023

### Artifact Evaluation Committee

- IEEE/ACM International Symposium on Microarchitecture 2023 (MICRO 2023, 2022)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2022)
- Conference on Machine Learning and Systems (MLSys 2020)
- Architectural Support for Programming Languages and Operating Systems (ASPLOS 2020)

---

## Teaching

ECE 757 (Advanced Computer Architecture II)

ECE 352 (Digital System Fundamentals)

ECE 5750 (High Performance Computer Architecture)

Spring 2024, Spring 2025

Fall 2024

Fall 2017